

TE Curve Verification Pipeline Familywise ONNX Report - residual_harmonic_lstm_sequence_dense240

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates exported ONNX models from the dataset input-mode retraining program. Each dataset/input-mode section uses dataset-matched held-out test curves and lists the exact model artifacts loaded from `models/`.

Rank-3 temporal ONNX exports are evaluated on the sequence-window test contract stored in each `training_config.snapshot.yaml`, including `sequence_length`, `sequence_stride`, `sequence_target_position`, and `maximum_sequences_per_curve`. The collage pages keep the measured TE trace at the original full-curve resolution; temporal ONNX predictions are overlaid at the evaluated sequence-target angles.

The report is diagnostic and family-specific. It does not replace an official multi-index model-promotion decision.

Output Artifacts

- output directory:
`output/validation_checks/track2_familywise_onnx_report/residual_harmonic_lstm_sequence_dense240/2026-07-10-09-24-21_track2_residual_harmonic_lstm_sequence_dense240_familywise_onnx_report ;`
- summary YAML:
`output/validation_checks/track2_familywise_onnx_report/residual_harmonic_lstm_sequence_dense240/2026-07-10-09-24-21_track2_residual_harmonic_lstm_sequence_dense240_familywise_onnx_report/track2_familywise_onnx_report_summary.yaml ;`
- model inventory CSV:
`output/validation_checks/track2_familywise_onnx_report/residual_harmonic_lstm_sequence_dense240/2026-07-10-09-24-21_track2_residual_harmonic_lstm_sequence_dense240_familywise_onnx_report/model_inventory.csv ;`
- per-curve metrics CSV:
`output/validation_checks/track2_familywise_onnx_report/residual_harmonic_lstm_sequence_dense240/2026-07-10-09-24-21_track2_residual_harmonic_lstm_sequence_dense240_familywise_onnx_report/per_curve_metrics.csv .`

Simplified Dataset + Setpoints

- dataset: `simplified_dataset` ;
- input mode: `setpoints` ;
- evaluated family: `residual_harmonic_lstm_sequence_dense240` ;
- dataset root: `data/simplified_dataset` .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	<code>te_residual_harmonic_lstm_sequence_dense240_fw_simplified_setpoints</code>	<code>2026-07-09-23-32-01_te_residual_harmonic_lstm_sequence_dense240_fw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
backward	<code>te_residual_harmonic_lstm_sequence_dense240_bw_simplified_setpoints</code>	<code>2026-07-09-23-53-53_te_residual_harmonic_lstm_sequence_dense240_bw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
global	<code>te_residual_harmonic_lstm_sequence_dense240_global_simplified_setpoints</code>	<code>2026-07-09-23-14-07_te_residual_harmonic_lstm_sequence_dense240_global_simplified_setpoints</code>	<code>simplified_curve_v1</code>

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	<code>models/simplified_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense240/forward/2026-07-09-23-32-01_te_residual_harmonic_lstm_sequence_dense240_fw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense240/forward/2026-07-09-23-32-01_te_residual_harmonic_lstm_sequence_dense240_fw_simplified_setpoints/python/residual_harmonic_lstm_sequence-epoch=061-val_mae=0.00356071.ckpt</code>
backward	<code>models/simplified_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense240/backward/2026-07-09-23-53-53_te_residual_harmonic_lstm_sequence_dense240_bw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense240/backward/2026-07-09-23-53-53_te_residual_harmonic_lstm_sequence_dense240_bw_simplified_setpoints/python/residual_harmonic_lstm_sequence-epoch=087-val_mae=0.00358639.ckpt</code>
global	<code>models/simplified_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense240/global/2026-07-09-23-14-07_te_residual_harmonic_lstm_sequence_dense240_global_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense240/global/2026-07-09-23-14-07_te_residual_harmonic_lstm_sequence_dense240_global_simplified_setpoints/python/residual_harmonic_lstm_sequence-epoch=042-val_mae=0.00360357.ckpt</code>

Aggregate Metrics

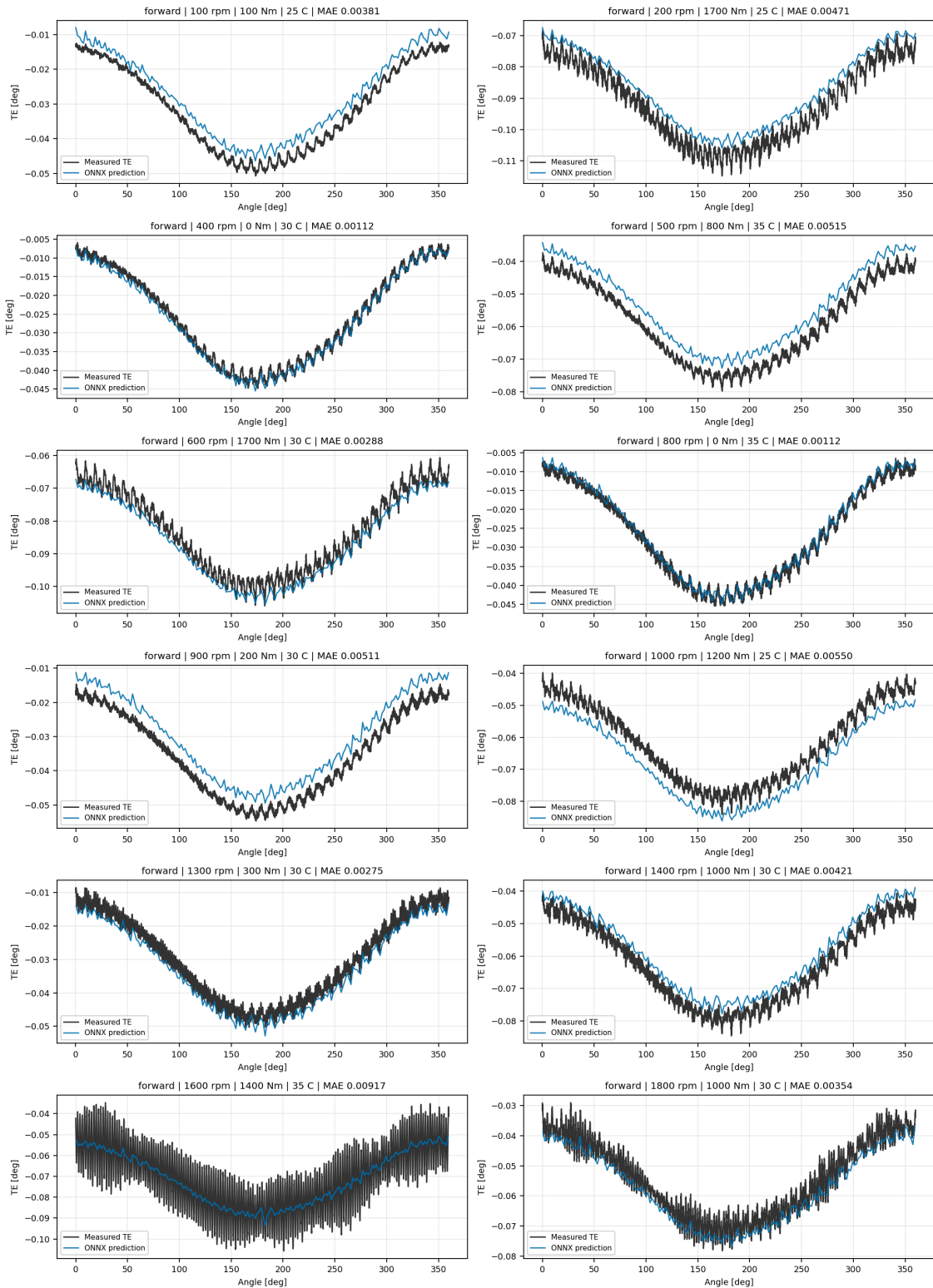
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.003230	0.003560	7.550	12.928
backward	97	0.003486	0.003874	8.093	13.883
global	194	0.003381	0.003750	7.878	14.033

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000025	0.002818	0.001404	0.003810
backward	0.000447	0.002811	0.001729	0.004132
global	0.000366	0.002832	0.001594	0.003848

Forward 12-Curve Page

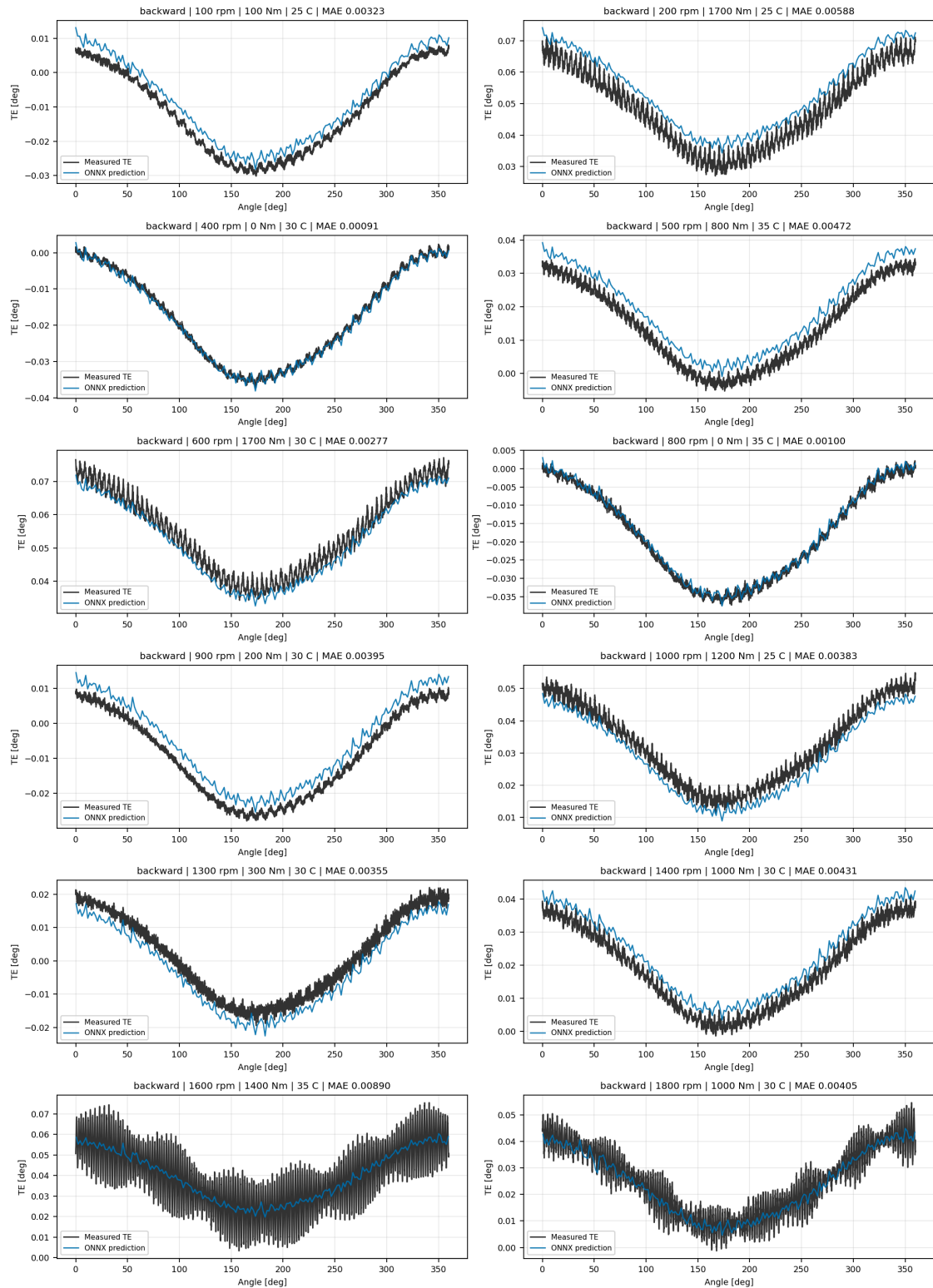
Simplified Dataset + Setpoints | residual_harmonic_lstm_sequence_dense240 | forward



simplified_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

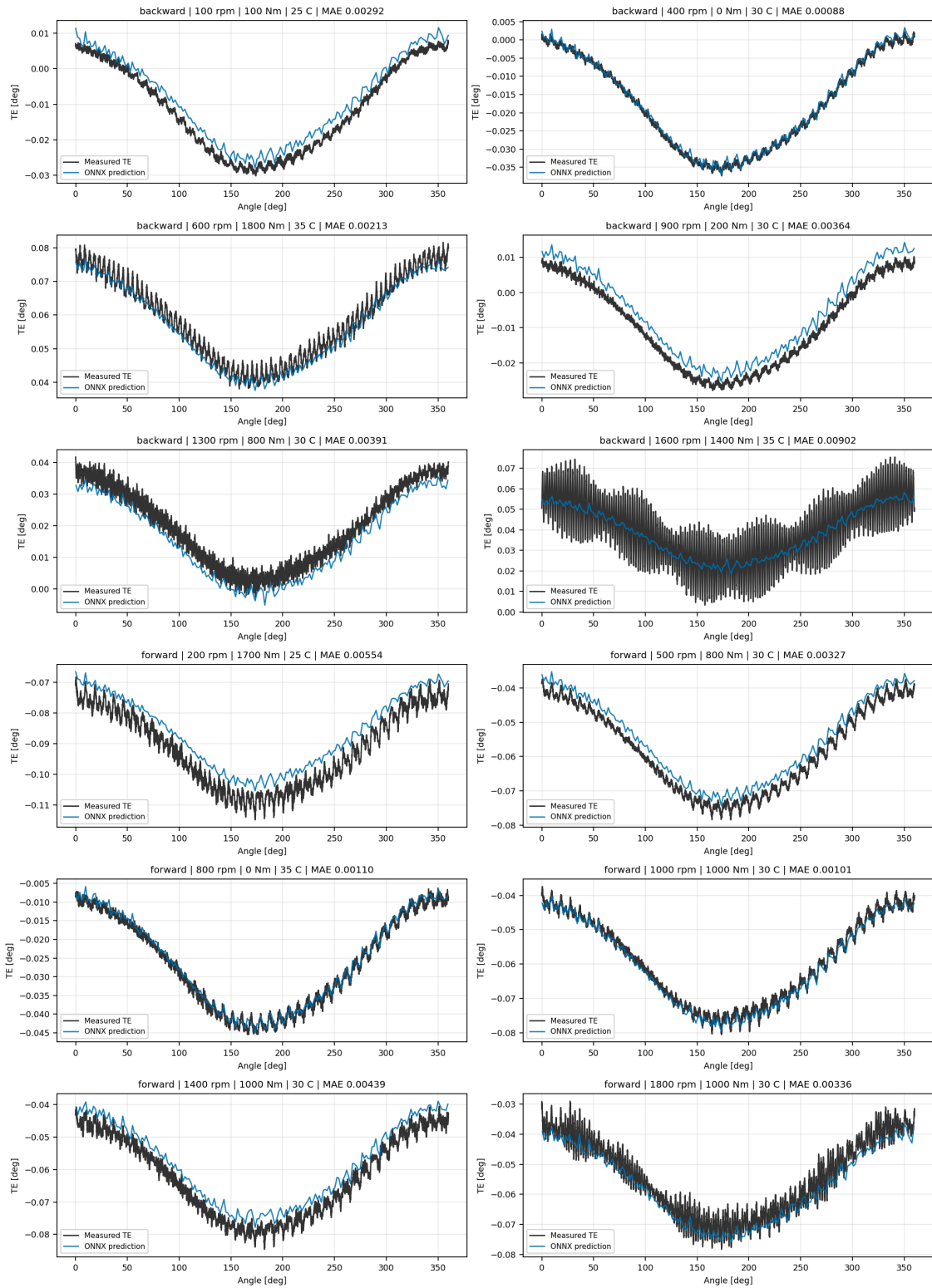
Simplified Dataset + Setpoints | residual_harmonic_lstm_sequence_dense240 | backward



simplified_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Simplified Dataset + Setpoints | residual_harmonic_lstm_sequence_dense240 | global



simplified_dataset_setpoints global 12-curve collage

Polished Dataset + Setpoints

- dataset: polished_dataset ;
- input mode: setpoints ;
- evaluated family: residual_harmonic_lstm_sequence_dense240 ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_residual_harmonic_lstm_sequence_dense240_fw_polished_setpoints	2026-07-10-00-54-03_te_residual_harmonic_lstm_sequence_dense240_fw_polished_setpoints	polished_setpoint_curve_v1
backward	te_residual_harmonic_lstm_sequence_dense240_bw_polished_setpoints	2026-07-10-01-19-43_te_residual_harmonic_lstm_sequence_dense240_bw_polished_setpoints	polished_setpoint_curve_v1
global	te_residual_harmonic_lstm_sequence_dense240_global_polished_setpoints	2026-07-10-00-33-07_te_residual_harmonic_lstm_sequence_dense240_global_polished_setpoints	polished_setpoint_curve_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense240/forward/2026-07-10-00-54-03_te_residual_harmonic_lstm_sequence_dense240_fw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense240/forward/2026-07-10-00-54-03_te_residual_harmonic_lstm_sequence_dense240_fw_polished_setpoints/python/residual_harmonic_lstm_sequence-epoch=067-val_mae=0.00199481.ckpt
backward	models/polished_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense240/backward/2026-07-10-01-19-43_te_residual_harmonic_lstm_sequence_dense240_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense240/backward/2026-07-10-01-19-43_te_residual_harmonic_lstm_sequence_dense240_bw_polished_setpoints/python/residual_harmonic_lstm_sequence-epoch=073-val_mae=0.00200521.ckpt
global	models/polished_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense240/global/2026-07-10-00-33-07_te_residual_harmonic_lstm_sequence_dense240_global_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/residual_harmonic_lstm_sequence_dense240/global/2026-07-10-00-33-07_te_residual_harmonic_lstm_sequence_dense240_global_polished_setpoints/python/residual_harmonic_lstm_sequence-epoch=045-val_mae=0.00200392.ckpt

Aggregate Metrics

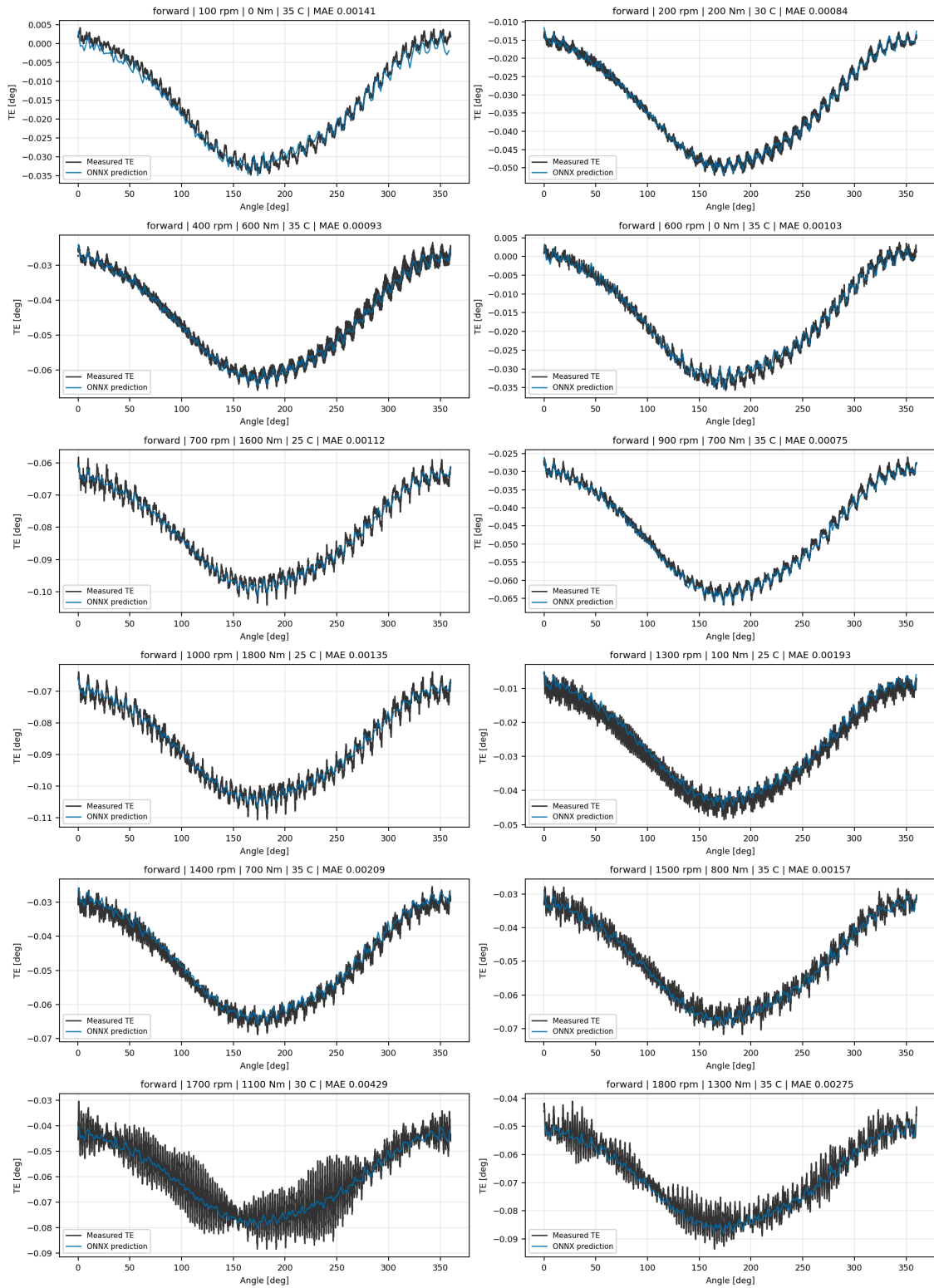
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.001961	0.002348	4.306	9.387
backward	94	0.002671	0.003144	4.984	10.865
global	194	0.002297	0.002733	4.612	10.970

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000030	0.000870	0.001669	0.003889
backward	-0.000247	0.001115	0.002223	0.006368
global	0.000232	0.000920	0.001958	0.005178

Forward 12-Curve Page

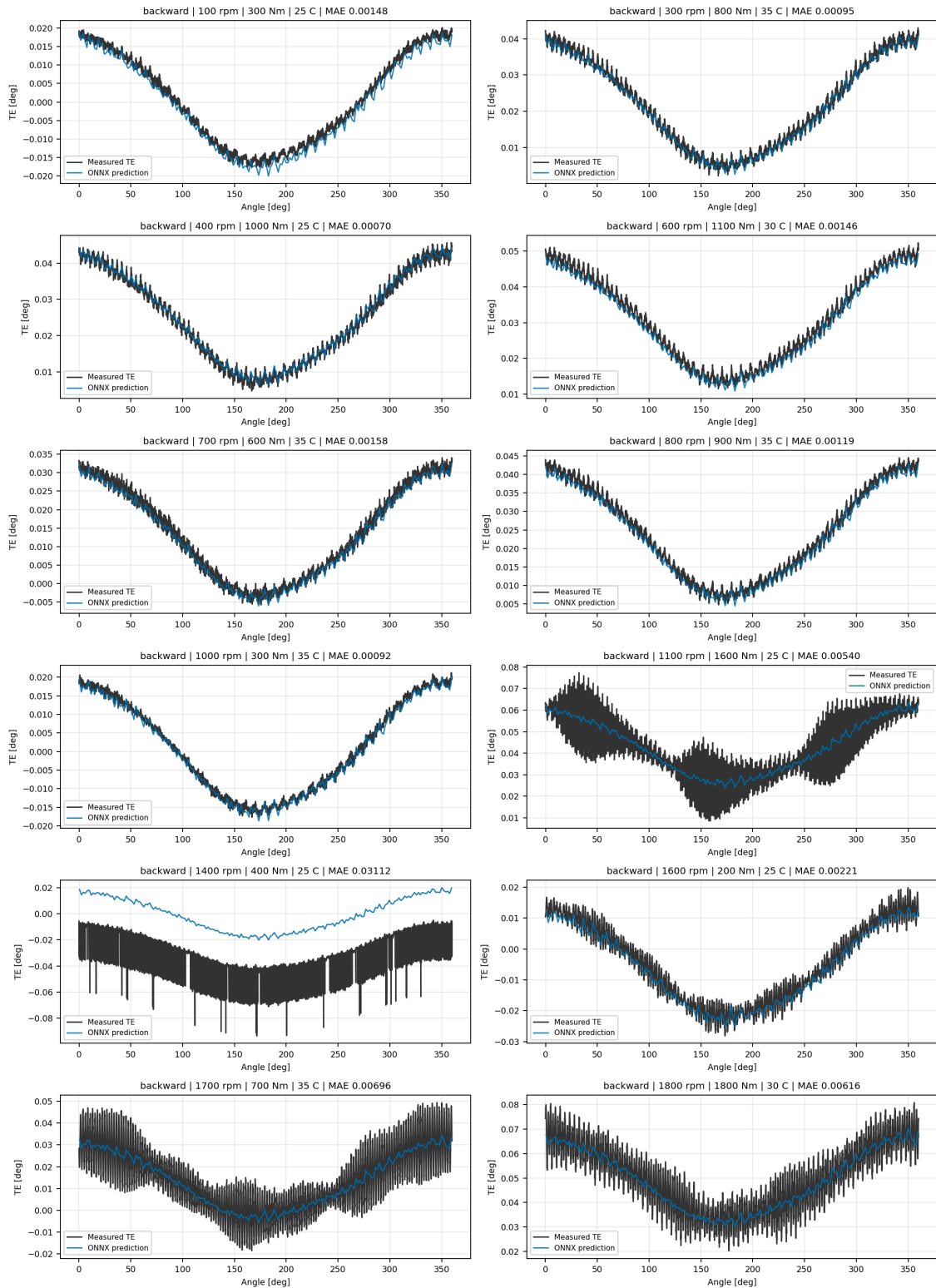
Polished Dataset + Setpoints | residual_harmonic_lstm_sequence_dense240 | forward



polished_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

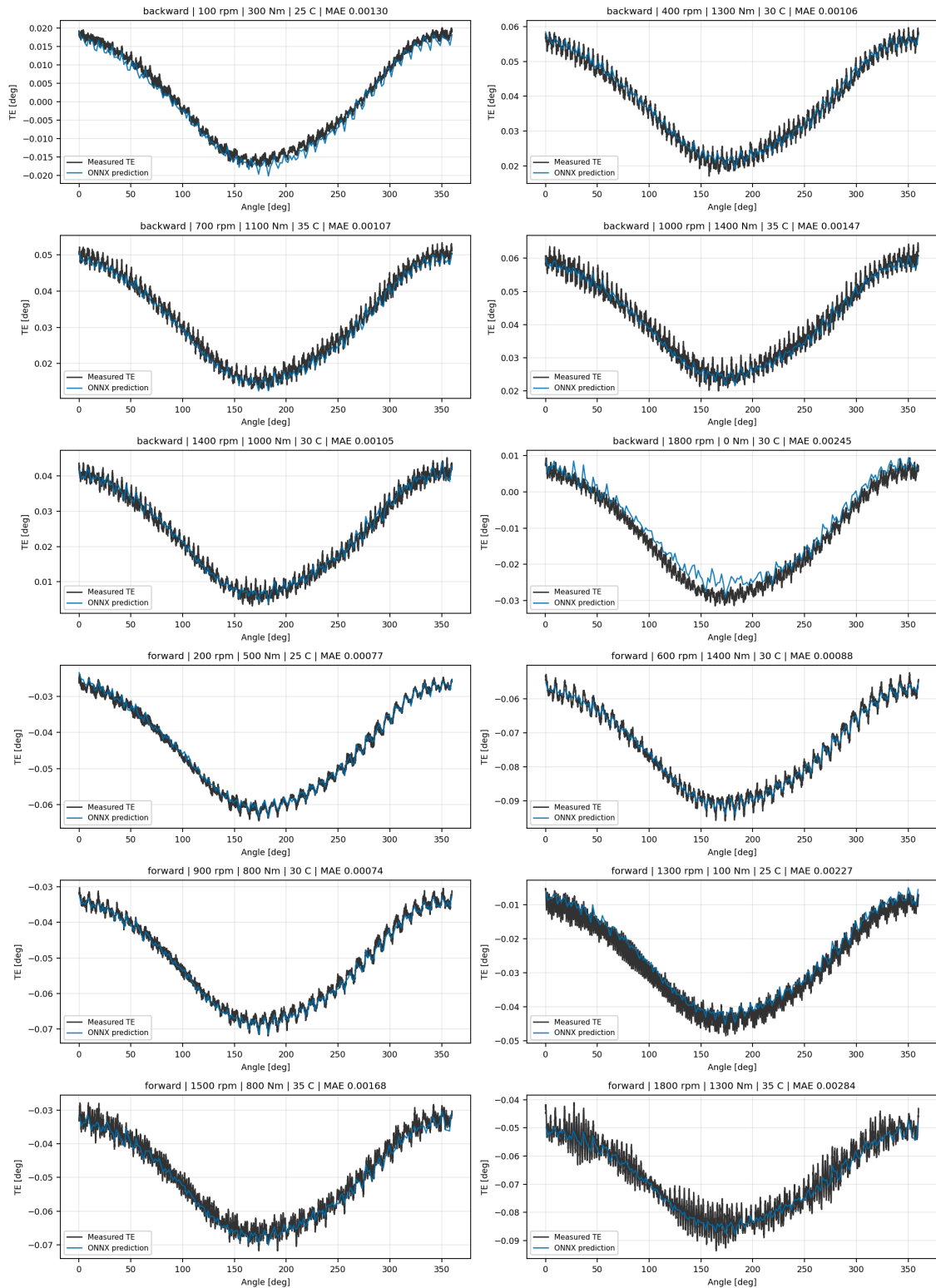
Polished Dataset + Setpoints | residual_harmonic_lstm_sequence_dense240 | backward



polished_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Setpoints | residual_harmonic_lstm_sequence_dense240 | global



polished_dataset_setpoints global 12-curve collage

Polished Dataset + Actual Values

- dataset: polished_dataset ;
- input mode: actual_values ;
- evaluated family: residual_harmonic_lstm_sequence_dense240 ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_residual_harmonic_lstm_sequence_dense240_fw_polished_actual_values	2026-07-10-02-31-07_te_residual_harmonic_lstm_sequence_dense240_fw_polished_actual_values	polished_point_v1
backward	te_residual_harmonic_lstm_sequence_dense240_bw_polished_actual_values	2026-07-10-02-54-22_te_residual_harmonic_lstm_sequence_dense240_bw_polished_actual_values	polished_point_v1
global	te_residual_harmonic_lstm_sequence_dense240_global_polished_actual_values	2026-07-10-02-07-46_te_residual_harmonic_lstm_sequence_dense240_global_polished_actual_values	polished_point_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/actual_values/exported/residual_harmonic_lstm_sequence_dense240/forward/2026-07-10-02-31-07_te_residual_harmonic_lstm_sequence_dense240_fw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/residual_harmonic_lstm_sequence_dense240/forward/2026-07-10-02-31-07_te_residual_harmonic_lstm_sequence_dense240_fw_polished_actual_values/python/residual_harmonic_lstm_sequence-epoch=055-val_mae=0.00202504.ckpt
backward	models/polished_dataset/actual_values/exported/residual_harmonic_lstm_sequence_dense240/backward/2026-07-10-02-54-22_te_residual_harmonic_lstm_sequence_dense240_bw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/residual_harmonic_lstm_sequence_dense240/backward/2026-07-10-02-54-22_te_residual_harmonic_lstm_sequence_dense240_bw_polished_actual_values/python/residual_harmonic_lstm_sequence-epoch=126-val_mae=0.00198495.ckpt
global	models/polished_dataset/actual_values/exported/residual_harmonic_lstm_sequence_dense240/global/2026-07-10-02-07-46_te_residual_harmonic_lstm_sequence_dense240_global_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/residual_harmonic_lstm_sequence_dense240/global/2026-07-10-02-07-46_te_residual_harmonic_lstm_sequence_dense240_global_polished_actual_values/python/residual_harmonic_lstm_sequence-epoch=055-val_mae=0.00202673.ckpt

Aggregate Metrics

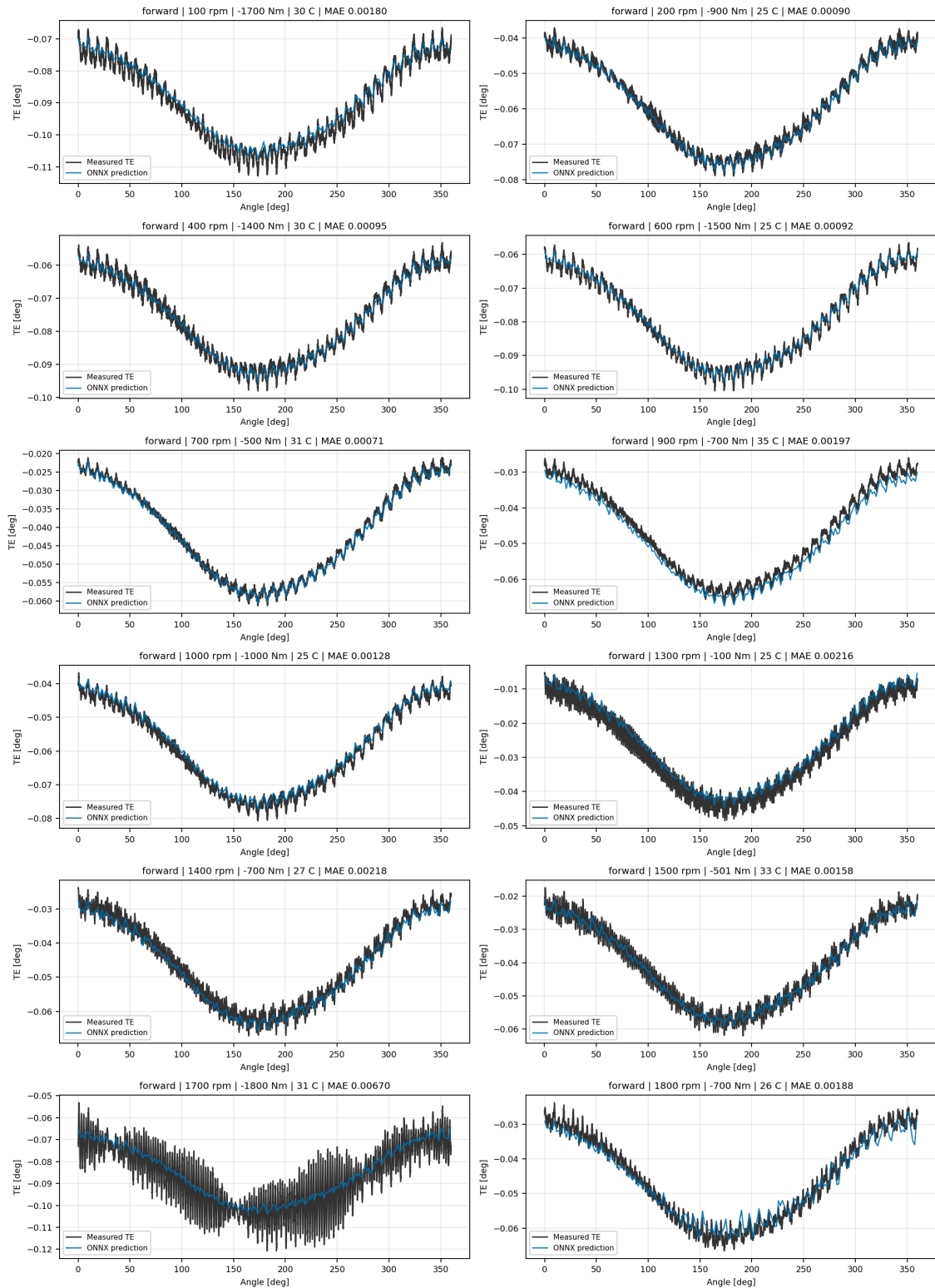
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.001962	0.002347	4.308	10.295
backward	94	0.002354	0.002851	4.508	10.769
global	194	0.002297	0.002725	4.613	10.984

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000435	0.000938	0.001630	0.004765
backward	0.000161	0.000501	0.002278	0.005896
global	-0.000091	0.000864	0.001961	0.005404

Forward 12-Curve Page

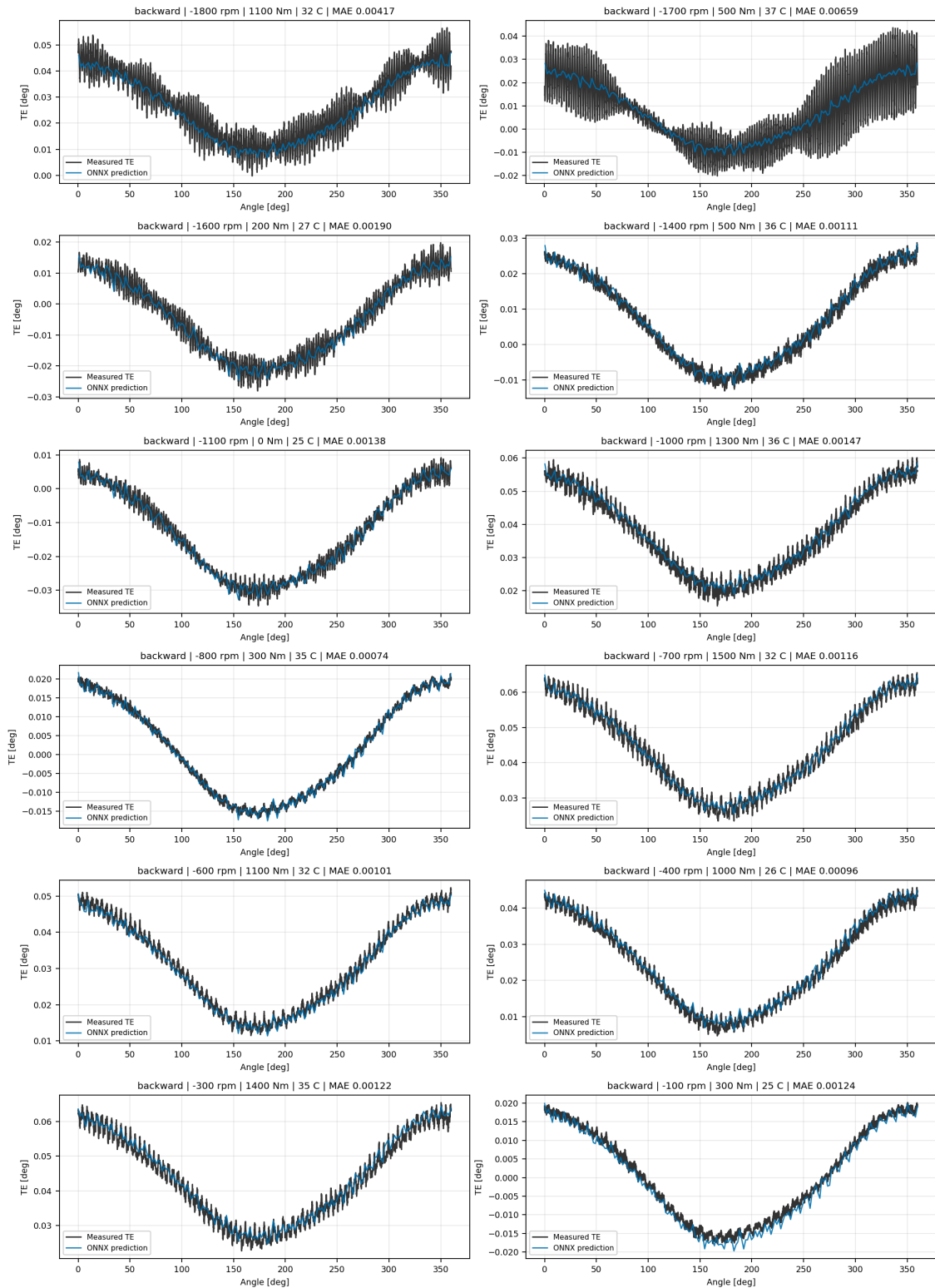
Polished Dataset + Actual Values | residual_harmonic_lstm_sequence_dense240 | forward



polished_dataset_actual_values forward 12-curve collage

Backward 12-Curve Page

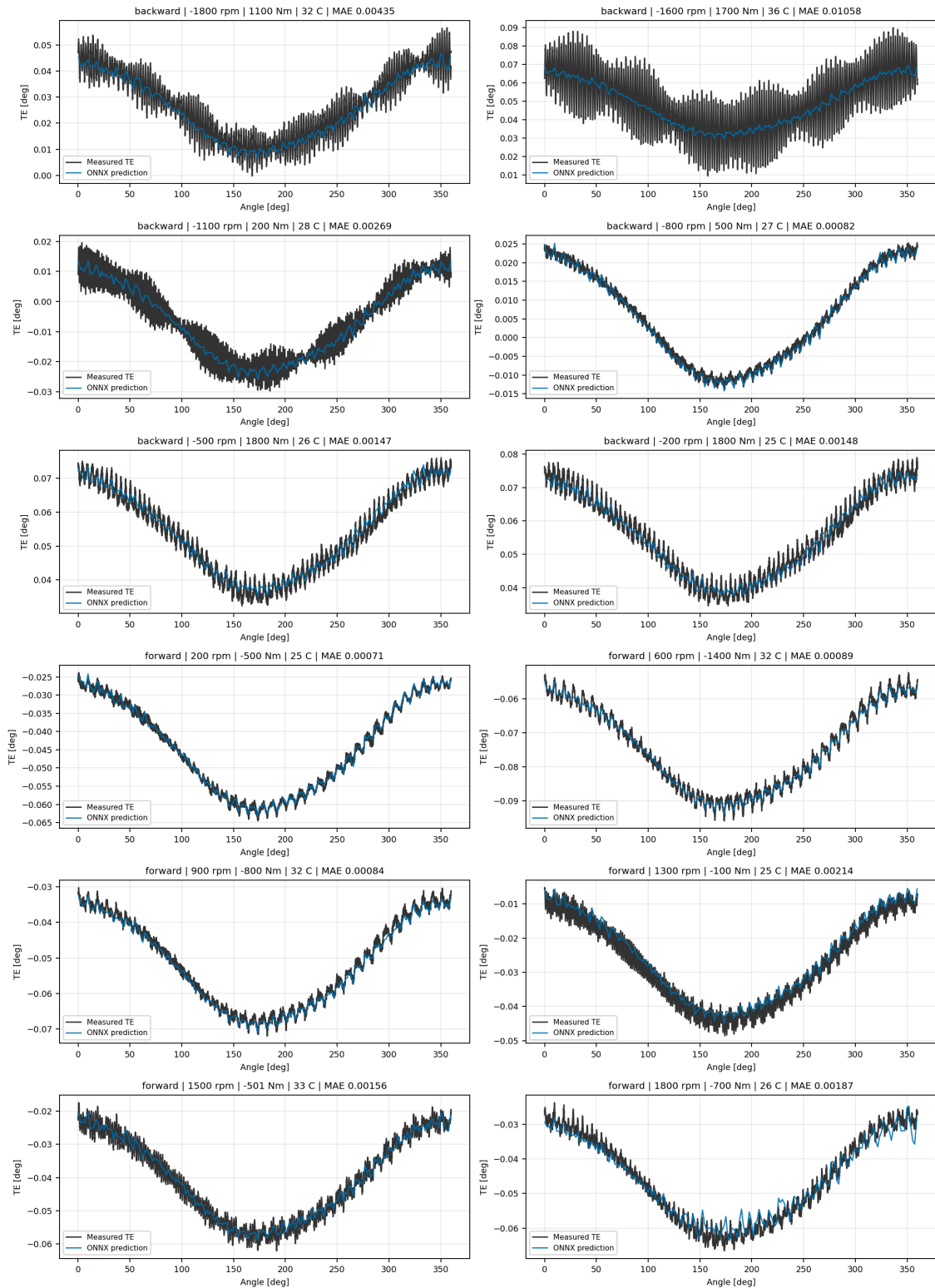
Polished Dataset + Actual Values | residual_harmonic_lstm_sequence_dense240 | backward



polished_dataset_actual_values backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Actual Values | residual_harmonic_lstm_sequence_dense240 | global



polished_dataset_actual_values global 12-curve collage